

TANMEET SINGH SACHDEVA

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SUMMARY

Mechatronics engineer who takes hardware from architecture to deployment. Specialises in power electronics, embedded firmware, and hardware verification, with end-to-end ownership of systems delivered to **Royal Australian Air Force**

EDUCATION

University of New South Wales, Sydney, Australia

Bachelor of Engineering – Mechatronic Engineering (**Distinction**, Honours)

Sep 2021 – Aug 2025

Sydney, NSW

PROFESSIONAL EXPERIENCE

Electronics Engineer – Cognitive Advantage

ESP32-S3, STM32, ESP-IDF, Altium, GNU Radio

Nov 2024 – Present

Canberra, ACT

- Delivered Control Board and Power Selector daughter board achieving **sub-2ms** power source switching and **sub-30µs** MOSFET switching, improving kit reliability and reducing manual intervention across deployments
- Achieved **97%** efficiency and $\pm 20\text{mV}$ p-p ripple on 12V/24V buck-boost supplies for deployable defence hardware
- Designed a capacitor based UPS with **sub-2ms failover** and 99% uptime, protecting Starlink, server, and mesh radio
- Reverse-engineered an undocumented **BLE** protocol, building a React Native mobile app and Raspberry Pi telemetry deployment for a COTS battery system, with the integrated solution delivered to the Royal Australian Air Force
- Programmed SDR using **GNU Radio** to scan **2–5GHz** spectrum, verifying frequency operation for 5G defence project
- Reduced wire harness validation time by up to **90%** through a modular **automated** test rig with automatic fault checking

Digital Accessibility Analyst – TPG Telecom

WCAG 2.2, User Research, Cross-functional Collaboration

Sep 2024 – Dec 2024

Sydney, NSW

- Improved digital accessibility for 10,000+ users via **WCAG 2.2** audits across 15 product areas, shaping design policy
- Presented full audit findings to **senior leadership**, aligning **design and engineering** teams across TPG, Felix, Lebara, and Vodafone on 5 prioritised remediation plans, driving organisation-level change for accessibility

ENGINEERING PROJECTS

Assistive Exoskeleton Control Systems – MATLAB/Simulink, CAN, HIL

Thesis, Control Systems Lead

Sep 2024 – Aug 2025

- Engineered state machine control logic validating hip and knee dynamics by testing software simulations on hardware
- Built an ESP32-S3 Hardware-in-the-Loop rig validating precise **PID** control at **1kHz** across 4 CubeMars actuators.
- Led a cross-functional team of 5 to integrate **12 sensors** and software systems in ensuring testing safety compliance

Sunswift Racing – CAN FD, STM32, 5G C-V2X

Embedded Systems Engineer

Feb 2024 – Nov 2024

- Architected an embedded network transmitting vehicle data across **CAN FD** from 12 STAR nodes, maximising reliability
- Developed Dash Connect, a **5G V2X** module delivering driver awareness at **sub-500ms latency** and **30fps** in real time
- Integrated **AWS IoT Core** and **Grafana** for live telemetry from Solar Car at **BWSC**, collaborating with **75** engineers

FATEH Motorsports (Formula Student) – HMI, Telemetry

Data Acquisition & Electronics Engineer

Oct 2021 – Dec 2022

- Designed HMI and telemetry systems using **LoRa** for real-time race data across **20+ sensors** improving trackside decisions
- Pitched an EV Formula Student business plan to industry judges, securing **3rd position** in PiEV at Formula Bharat 2022
- Achieved **1st/50** teams at Formula Bharat for **engineering design** and **business use case** strength across India

TECHNICAL SKILLS

Embedded & Firmware: C++, Python, ESP32-S3, STM32, ESP-IDF, FreeRTOS, HIL Testing, AI Automation

Power Electronics & PCB: Altium, Buck-Boost Converters, Power Supply Design, MOSFETs, Hardware Bring-up

Communications & Test: CAN FD, FlexRay, BLE, 5G C-V2X, SDR, GNU Radio, Oscilloscopes, MATLAB/Simulink, Linux

LEADERSHIP & ACHIEVEMENTS

- Young Engineers Australia Committee** – organised a site visit to **BESS**, and industry panel discussion
- Sunswift Racing** – led 2 electronics projects from concept through to final implementation and deployment
- European Defence Club Hackathon** – built and presented a drone detection algorithm in front of judges from defence

Formula Bharat **1st/50** teams | Australia Global University Award (AUD 10,000) | UNSW-Ti Articulation Scholarship (AUD 10,000)